

The circuit shown in Figure 3 shows a single switch forward converter. Winding N_1 has 50 turns and has an inductance of 5 mH. The supply voltage V_s is 240 volts, and the switching frequency f_s is 60kHz. The converter is being designed to supply a maximum of 60 volts at a duty cycle of 0.75.

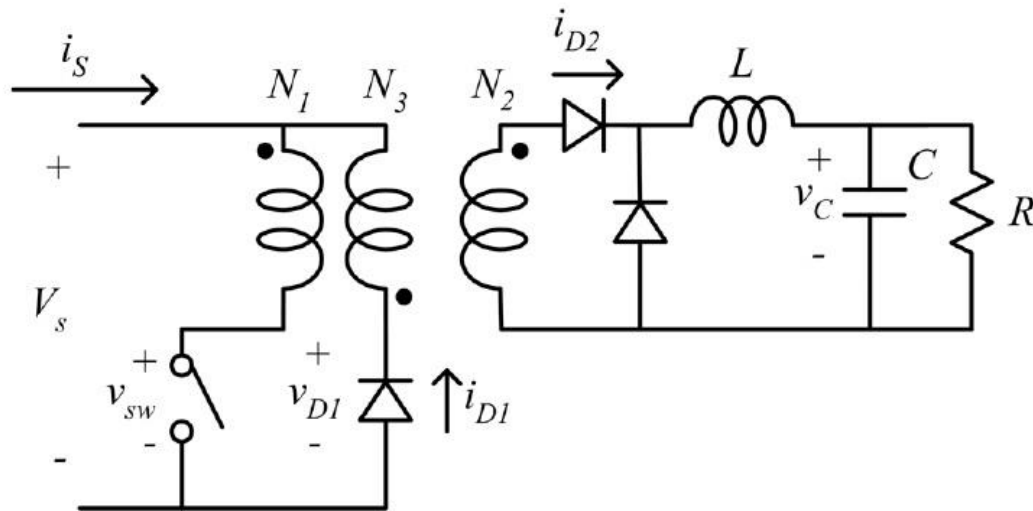


Figure 3. Single switch forward converter circuit

- Sketch waveforms, including their alignment in time, showing i_s , i_{D1} , v_{D1} and v_{sw} .
- Calculate a suitable number of turns for winding N_2 .
- Calculate a suitable number of turns for winding N_3 .
- Calculate the peak switch voltage v_{sw} for this circuit.
- Discuss the advantages and disadvantages of this circuit compared to a fly-back converter.